

BLAST FURNACE PRODUCTION SECTION												ON-DT	ON-Month	
DATE:	1. Apr. 2024			DAILY PARAMETERS REPORT				EB		0	Crude steel To HM Ratio		0.967	0.967
Monthly Plan (T)		ABP	OPP	Sinter Prod. / Nut Coke				EB		0	STOCKS		ON-DT	Prev.Day
Daily Rate (T/Day)		16774	Sinter Prod. / Nut Coke				EB		0	Sinter stock		High lines (1-7)	1300	700
H.M.SENT TO SMS's		15802	Sinter Prod. / Nut Coke				EB		0	I/Ore lump stock		Em.Storage	10400	9000
			Sinter Prod. / Nut Coke				EB		0			OHP	3052	3702
			Sinter Prod. / Nut Coke				EB		0			OHP # 2	14330	16266
			Sinter Prod. / Nut Coke				EB		0			BF	35097	35229
			Sinter Prod. / Nut Coke				EB		0			CO	1814	3365
			Sinter Prod. / Nut Coke				EB		0			BFs	0	0
			Sinter Prod. / Nut Coke				EB		0			Coke stock supply from stock	369	755
			Sinter Prod. / Nut Coke				EB		0			CDI STOCK (1 - 7)	455	455
			Sinter Prod. / Nut Coke				EB		0			CDI STOCK	461	415
			Sinter Prod. / Nut Coke				EB		0			SHIPPING REPORT	369	755
			Sinter Prod. / Nut Coke				EB		0			HM (PCM)	0	465
			Sinter Prod. / Nut Coke				EB		0			PROD.	900.36	305.65
			Sinter Prod. / Nut Coke				EB		0			DESP.	900.36	305.65
			Sinter Prod. / Nut Coke				EB		0			STOCK	6092.63	
			Sinter Prod. / Nut Coke				EB		0			GRANULATED SLAG		
			Sinter Prod. / Nut Coke				EB		0			BY ROAD		
			Sinter Prod. / Nut Coke				EB		0			BY RAIL		
			Sinter Prod. / Nut Coke				EB		0			Total Desp.	2573.65	2573.65
			Sinter Prod. / Nut Coke				EB		0			STOCK	2384535.94	
			Sinter Prod. / Nut Coke				EB		0			Asking Rt.Of H.M.Prod.		
			Sinter Prod. / Nut Coke				EB		0			18042		
			Sinter Prod. / Nut Coke				EB		0					
			Sinter Prod. / Nut Coke				EB		0					
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			Sinter Prod. / Nut Coke				EB		0					
			Sinter Prod. / Nut Coke				EB		0					

HOT METAL AND SLAG QUALITY

Parameter	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8	SHOP	TILL
Avg. % 'Si'	0.71	0.92	0.86		0.99	0.75	0.84	0.84
Min/Max	0.40/1.19	0.69/1.50	0.53/1.06		0.73/1.19	0.49/1.03		
Avg. % 'S'	0.043	0.038	0.036		0.023	0.038	0.035	0.035
Min/Max	0.030/0.062	0.020/0.062	0.030/0.052		0.020/0.028	0.030/0.050		
Avg. % MgO	7.18	7.63	7.18		7.24	6.32	7.11	
Min/Max	6.59/7.51	7.1/8.6	7.05/7.36		6.82/7.42	5.93/6.69		
Avg. % Al ₂ O ₃	17.57	17.46	18.12		17.33	19.38	17.97	
Avg. % FeO	0.75	0.73	0.7		0.67	1.15	0.80	
Avg. % K ₂ O	0.6	0.58	0.38		0.37	0.49	0.48	
BASICITY(-)	0.87	0.91	0.92		0.95	0.94	0.92	
Min/Max	0.82/0.95	0.74/0.98	0.88/0.98		0.94/1.01	0.88/1.05		
Avg. % 'P'	0.11	0.11	0.11		0.10	0.10	0.11	
Min/Max	0.10/0.13	0.10/0.13	0.10/0.13		0.10/0.12	0.08/0.11		

Fce.No.	Si→1.00	Si→0.85	No.of Casts	Tap to Tap >1:00 Hrs
1	1	1	9	3
4	2	3	7	2
5	1	4	9	0
6			0	
7	3	6	11	0
SHOP	7	14	36	5

H ₂ IN TOP GAS				
	BF-1	BF-4	BF-5	BF-7
G	4.60	4.80	4.20	STOP

CO/CO ₂ RATIO				
G	1.23	1.20	1.29	1.21
% Gas Util.				
G	44.85	45.5	43.75	45.19

MIXER	%Si	%Mn	%Sul	%P
Mixer-III	0.84	0.09	0.036	0.11
Mixer-IV	0.79	0.08	0.051	0.10

PELLET CHEMICAL AND SEIVE ANALYSIS									
BF # 8 =>	Fe	SiO ₂	Al ₂ O ₃	CaO	MgO	+ 10 MM	- 5 MM	M.Size	Basicity
BF # 7 =>						62.90	2.00	10.70	Slag Comp.
									#DIV/0!
									0.00
									#DIV/0!
									0.00

SKIP IRON ORE			
BF # 1	Fe (%)	SiO ₂ (%)	Al ₂ O ₃ (%)
BF # 1	62.39	5.04	3.19
BF # 4	62.56	3.33	4.78
BF # 5	63.47	3.60	3.25
BF # 7	61.69	5.89	3.63
BF # 8	61.04	7.06	3.12
Avg.	61.04	7.06	3.12

BF SKIP IORE SEIVE ANALYSIS			
Fce.No.	+40 mm	- 10 mm	M.Size
BF # 1	23.50	13.30	27.60
BF # 4	25.90	13.90	28.40
BF # 5	25.30	13.00	27.80
BF # 7	18.80	14.10	26.40
BF # 8	19.20	14.00	25.80
AVG.	22.54	13.66	27.20

% FINES IN BF SKIP SINTER				
Fce.No.	-10 mm	- 5 mm	M.Size	TOTAL Fe
BF # 1	37.10	7.10	15.60	54.04
BF # 4	41.10	7.20	15.20	51.92
BF # 5	37.60	7.60	16.10	52.37
BF # 7	39.90	7.90	15.20	52.76
BF # 8	36.70	10.00	14.70	
AVG.	38.48	7.96	15.36	

% FINES IN BF SKIP COKE			
Fce.No.	-40 mm	- 25 mm	M.Size
BF # 1	32.30	1.60	52.30
BF # 4	35.30	2.20	51.10
BF # 5	32.50	1.90	52.40
BF # 7	33.00	1.30	52.40
BF # 8	41.30	1.30	43.70
AVG.	34.88	1.66	50.38

COKE QUALITY									
PROXIMATE ANALYSIS					COLD STRENGTH			HOT STRENGTH	
	Moisture	Ash	V.M	F.C	M-40	M-10	Sulphur	CSR	CRI
CSP-I: A	3.6	14.6	0.9	84.1			0.59		
CSP-II: A	3.9	15.0	0.9	83.6	80.2	6.8	0.61	64.1	23.9
CSP-III: A	4.6	15.0	0.8	83.6			0.55		
CSP-IV: A	0.4	14.8	0.9	83.8	88.2	6.2	0.51		
BF # 8 STOCK HOUSE SURFACE MIX.COKE =>								65.9	22.7

COAL BLEND (%)												
	DATE	PCC	MCC	Aus(H)	BENGA	LVHR (H)	TUHU(Phard)	Russian (S)	AUS(S)	USA(H)	USA(S)	% IMP
CHP - I	1-Feb-24	13.0	2.0	38.0	9.0	5.0	0.0	0.0	12.0	11.0	10.0	85.0
CPP - II	1-Feb-24	13.0	2.0	38.0	9.0	5.0	0.0	12.0	10.0	0.0	11.0	85.0
												CHP - I
												11.2
												CPP - II
												11.9

SINTER CHEMICAL ANALYSIS

SINTER PLANT-2												
NORM(Basicity)				(2.05 - 2.10) (14-11-23) Nut Coke				21 Kg/T of Sinter				
NORM(Mgo)				(2.00 - 2.50) (06-11-23) MILL SCALE				23 Kg/T of Sinter				
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component	
AO	50.34	9.27	8.61	1.48	15.35	2.18	1.78	6.74	OUT	1.78	26.47	
A1	49.46	9.27	7.29	2.07	17.38	2.64	2.38	10.09	OUT	2.38	C I (SP # II)	
A2	51.25	9.27	7.52	1.73	15.33	2.29	2.04	7.81	OUT	2.04	Coke	Flux
BO	52.56	9.25	6.54	1.24	15.06	1.93	2.30	8.52	OUT	2.30	91.8	DTI
B1	51.48	9.25	7.03	1.14	15.98	2.31	2.27	8.95	OUT	2.27	92.1	76.5
B2	53.63	9.25	6.10	1.07	14.09	1.92	2.31	7.99	OUT	2.31		
CO	50.64	9.25	7.39	1.05	16.42	2.34	2.22	9.03	OUT	2.22		
C1												
C2	51.50	9.25	7.20	1.65	15.29	2.12	2.12	8.09	OUT	2.12		
DAY AVG.	51.36	9.26	7.21	1.43	15.61	2.22	2.18	8.40	OUT	2.17		

SINTER PLANT-3												
NORM(Basicity)				(1.80 - 1.85) (03-02-24) Nut Coke				0 Kg/T of Sinter				
NORM(Mgo)				(2.00 - 2.50) (06-11-23) MILL SCALE				12 Kg/T of Sinter				
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component	
AO	51.99	9.15	7.37	1.36	14.22	2.81	1.93	6.85	OUT	1.93	23.64	DTI
A1	54.47	9.15	6.52	1.22	12.57	2.23	1.93	6.05	OUT	1.93	79.5	
A2	53.96	9.15	6.58	1.26	13.04	2.36	1.98	6.46	OUT	1.98		
BO	52.66	9.21	7.09	1.30	13.91	2.67	1.96	6.82	OUT	1.96		
B1	52.48	9.21	6.84	1.21	14.12	2.94	2.06	7.28	OUT	2.06		
B2	53.07	9.21	6.91	1.34	13.63	2.48	1.97	6.72	OUT	1.97		
CO	55.54	9.21	5.88	1.42	11.48	1.96	1.95	5.60	OUT	1.95		
C1												
C2	54.29	9.21	5.43	0.95	13.60	2.45	2.51	8.17	OUT	2.50		
DAY AVG.	53.56	9.19	6.58	1.26	13.32	2.49	2.04	6.74	OUT	2.03	52.72	

Sample ID											Slag Comp	
	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	23.92	53.40
SP-3/S/MC/2A0	52.61	9.60	6.69	1.53	14.20	2.68	2.12	7.51	OUT	2.12	24.21	
SP-3/S/MC/2A1	51.60	9.60	7.33	1.91	14.58	2.71	1.99	7.25	OUT	1.99		
SP-3/S/MC/2A2	52.47	9.60	7.01	1.49	14.13	2.73	2.02	7.12	OUT	2.02	C I (SP # III)	
SP-3/S/MC-2/B0	54.58	9.71	6.98	0.93	12.18	2.16	1.75	5.20	OUT	1.74	Coke	Flux
SP-3/S/MC-2/B1	53.81	9.71	6.49	1.33	13.08	2.54	2.01	6.59	OUT	2.02	86.4	DTI
SP-3/S/MC-2/B2	53.16	9.71	7.03	1.38	13.15	2.76	1.87	6.12	OUT	1.87	85.9	
SP-3/S/MC-2/C0	53.68	9.71	6.80	1.49	13.11	2.30	1.93	6.31	OUT	1.93		
SP-3/S/MC-2/C1												
SP-3/S/MC-2/C2	54.03	9.71	6.34	1.43	12.91	2.26	2.04	6.57	OUT	2.04		
DAY AVG.	53.24	9.67	6.83	1.44	13.42	2.52	1.97	6.58	OUT	1.96		

F.N	Duration		REASONS	ST	LB	RB	FB	Down Time	Furnace No	1	4	5	6	7	8	TOTAL
I	05:00	06:00	N ₂ P ₂ RESSURE LOW			1:00			ABP	0	77000	77000	0	118000	242000	514000
						1:00	23:00:00	0:15:00	OPP	29500	72000	82700	0	117800	212000	514000
IV			==NIL==						M/Rate	44070	78720	66870	0	103170	210390	503220
							24:00:00	0:00:00	Weekly Plan	1900	2850	2800	0	4000	8400	19950
V			==NIL==						Prod.(Cumm.)	1469	2624	2229	0	3439	7013	16774
							24:00:00		Fullfillment	77.3	92.1	79.6	#DIV/0!	86.0	83.5	84.1
VI			CAPTIAL REPAIR CAT - I						Today'sPlan	1900	2850	2800	0	4000	8400	19950
									HM Prod	1469	2624	2229	0	3439	7013	16774
VII	06:00	05:59	POOR WIND ACCEPTANCE			23:59			Cu.Prod	1469	2624	2229	0	3439	7013	16774
						23:59	0:01:00		D/Rate	1469	2624	2229	0	3439	7013	16774
VIII	02:00	04:00	PCB TRIPPED DUE TO N2 PRESSURE LOW			2:00			No. OF SKIPS (6:00 TO 6:00)							
	04:00	05:00	TO DRY THE FCE.		1:00				A	194	210	206		252	42	862
	05:00	06:00	FCE. ON SCHD. MAINT. DUE TO N2 PRESSURE LOW		1:00				B	214	218	190		236	44	858
					1:00	1:00	2:00	2:00:00	C	225	236	230		278	36	969
			TOTAL	1:00:00	1:00:00	26:59:00	91:01:00	8:14:45	Total skips	633	664	626	0	766	122	2689

COAL DUST INJ. (Kg/THM)				
BF				
No.	DAILY/ Date	CDI Rate(kg/thm	HM Production on that date(T)	CDI in tonnage(T)
1	25.03.2021	124	1852	230.15
4	14.12.2022	226	1250	281.90
5	22.11.2022	157	2000	314.92
6	10.10.2022	172	1821	313.14
7	11.12.2022	149	3526	523.83
8	05.11.22	192	7651	1472.00
SHOP	12.10.22	157	13701	2151.06

CDI On Tonnage Basis			
BF			
No.	DAILY	Best CDI On tonnage Basis(T)	HM Production on that date(T)
1	25.03.2021	230.15	1852
4	20.10.2022	356.00	2568
5	30.07.2022	371.30	2676
6	19.09.2022	334.99	2044
7	10.12.2022	540.30	4176
8	30.11.2022	1575.97	8793
SHOP	25.01.2023	2721.79	18771

Annual ABP Target(FY 2023-24)			Till Date
HM Target(MT)	6.20	0.02	
Coke RATE(kg/thm)	430	458	
CDI Rate(kg/thm)	125	106	
FUEL RATE(kg/thm)	587	595	
Financial Data			
Cost of HM Production in of FY 2022- 23(per ton of HM)	Rs 38648		
Cost of HM Production till Dec'23(per ton of HM)	Rs 35723		
Cost of HM Production in Dec'23(per ton of HM)	Rs 36230		
Contribution Margin per ton (Profit) for Pig Iron in Dec'23	Rs 1826		
Cost of Skip coke Till Dec'23 (per ton)	Rs 41363		
Cost of Skip Coke In Dec'23(per ton)	Rs 45682		
Previous Year ABP vs Actual (FY2022-23)	ABP	ACTUAL	
HOT METAL(MT)	6.40	5.42	
COKE RATE(kg/thm)	410	448	
CDI RATE(kg/thm)	115	120	
Fuel Rate(kg/thm)	557	585	
Pig Iron Produced till DEC'23	51189 T		
Pig Iron Produced in DEC'23	700 T		

Stoppage on Date					1:00:00	01.00.00
Low Blast on Dt					1:00:00	01.00.00
Reduced blast on Dt	1:00			23:59	2:00:00	02.59.00
No of Tappings	9	7	9	0	11	47
No of Not Dry casts	0	0	0	0	0	0
Coke Rate On Date	477	475	468		510	458
Coke Rt for month	477	475	468		510	458
Hot Blast Temp.	812	1002	997		953	1094

Metal ladle position			Slag ladle position		
In circulation		+80	Total	In Use	Under
Open Hearth		+75			M/Maint.
Reserve		+70	32	32	
U/Repair		-70	Total No of Hot Metal loads		
Cooling		Total	0	No.of Tapedo	
Mixer Position & M- III		4	338	% Screen Loss	Loads sent to OH
Pouring AT M-IV		5	402		Loads sent to CS
6:00:00 AM SMS-III (TPD)		3	857		Loads sent to PCM
					Loads sent to FDY
%SINTER FINES	SP # II	SP # III	% COKE FINES	BF#7	BF#8
	16.89	20.96		0.00	0.00
Coal Dust Injection	BF#1	BF#4	BF#5	BF#6	BF#7
Coal Consum.(T)	178.32	217.70	225.21	0.00	209.32
Coal Rate KG/THM	121	83	101	#DIV/0!	61
Coal Prog. KG/THM	11	13	13	0	12

Raw Mat. Position						
(FROM RMD)	L/STONE	Mn.Ore	Qz.	Conv.slagn	CDI COAL	Pellet (OHP)
Receipt	0	0	0	0	918	0
Stock	4625	169	2005	1500	1131	3124

RECORDS						
B.Fces.	Hot Metal Production		COKE RATE		CDI RATE	
	MONTHLY	MONTHLY	MONTHLY	MONTHLY	MONTHLY	MONTHLY
BF # 1	61574	AUG.'17	436	MAR.'12	124	JAN.'24
BF # 4	80225	JAN.'24	438	MAR.'23	130	DEC.'22
BF # 5	82758	MAR.'22	418	FEB.'09	122	NOV.'22
BF # 6	89535	JAN.'09	397	JAN.'02	144	OCT.'22
BF # 7	128247	OCT.'17	428	MAR.'19	124	NOV.'22
BF # 8	240960	MAR.'21	370	SEP.'21	160	DEC.'23
SHOP	554334	JAN.'23	414	SEP.'21	138	OCT.'22
PCM POURING	41	2451	30-06-2023			